Behavior-Aware Multilingual Voice-Based Local Service Discovery and Workforce Empowerment System Using a Multi-Agent AI Framework

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Project Proposal Report

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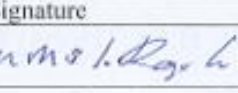
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid growth of digital service platforms has transformed the way individuals discover and access local services. However, many existing systems still struggle to support real-world communication patterns, particularly in multilingual environments such as Sri Lanka where users frequently switch between Sinhala, Tamil, and English during everyday conversations. In addition, users often provide incomplete or fragmented speech when requesting urgent services, making it difficult for traditional service discovery systems and voice assistants to accurately understand user intent.

This research proposes the design and evaluation of a behavior-aware multilingual voice-based local service discovery system using a multi-agent artificial intelligence framework. The system aims to improve service accessibility, reliability, and response efficiency by enabling users to request services through natural voice interactions while supporting mixed-language communication and urgency-sensitive requests.

The proposed solution consists of several intelligent components. A multilingual voice intent model interprets spoken commands containing language switching and incomplete phrases while detecting urgency levels using acoustic speech features such as pitch variation, speech rate, and energy patterns. A cooperative multi-agent coordination framework then identifies and ranks the most suitable service providers by analyzing contextual factors including location, travel distance, provider availability, reliability, and urgency level. In addition, an AI-driven workforce intelligence framework promotes fair job distribution, provider reputation management, and predictive workforce demand analysis to improve employment sustainability for service providers.

The system also integrates a behavior-aware interaction layer that enhances transparency and user experience by providing success prediction, security mechanisms, and intelligent guidance before service requests are confirmed. By combining voice intelligence, multi-agent coordination, and workforce analytics, the proposed system aims to create a smarter, more reliable, and accessible local service ecosystem.

Keywords: Multilingual Voice Interaction, Multi-Agent Systems, Service Discovery, Behavior-Aware AI, Workforce Intelligence

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
CNN	Convolutional Neural Network
ETA	Estimated Time of Arrival
MFCC	Mel Frequency Cepstral Coefficients
NLP	Natural Language Processing
SDG	Sustainable Development Goals
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure
MAS	Multi-Agent System
DB	Database
GPS	Global Positioning System
REST	Representational State Transfer
JSON	JavaScript Object Notation
API-GW	API Gateway

Table 1:LIST OF ABBREVIATIONS

LIST OF APPENDICES

Appendix Title	Appendix Title
Appendix A	**Survey Instruments**
Appendix B	**Detailed System Diagrams**
Appendix C	**Preliminary UI Mockups**
Appendix D	**Risk Analysis**
Appendix E	**Ethical Documents**

Table 2 :LIST OF APPENDICES

1. INTRODUCTION

1.1. Background and Context

The fast development of the on-demand online service platforms has changed the way people have learned to find and receive services like home repairs, healthcare, transportation, and emergency services. Nevertheless, it is still a great challenge to find the most fitting service provider at any time as the situation of the user, service presence, and the environment is dynamic. Services are often needed by the users depending on the numerous contextual parameters such as location, urgency, time of travel, and reliability of the provider. The traditional service discovery systems are often based on search mechanisms or simple ranking models which do not support dynamic response to the contextual differences introduced by real time situations and hence the recommendations provided are not always relevant to the user's needs. The problem impacts a large user base, especially those living in cities and who require urgent services using digital platforms. Despite the ever-growing economy of on-demand services worldwide, most of the available service recommendation systems have a hard time combining the contextual aspects of time sensitivity, geographic distance, and provider availability in the same time. As a result, the need to have intelligent decision-making mechanisms that are capable of dynamically adapting to changing user contexts is on an upward trend. The existing service discovery solutions mostly include centralized recommendation algorithms or rule of thumb filtering algorithms with little contextual awareness. The methods normally use standard ranking features like the rating of providers or pre-defined filters, which cannot be used in managing dynamic situations such as an emergency, changes in traffic patterns, or the availability of services. Moreover, most systems consider service discovery a one-time decision-making process but not a process that distributes responsibilities among specialized parts. The current developments in artificial intelligence show that multi-agent systems can improve complex decision-making by allowing various specialized agents to cooperate and can change strategies in uncertain settings. In spite of these developments, current multi-agent strategies usually do not have adaptive coordination mechanisms, which can integrate contextual reasoning with dynamic ranking strategies. This shortcoming identifies a gap in research on how to develop intelligent coordination models that can combine various contextual elements during the recommendation of service providers. To overcome this obstacle, in this study, the Adaptive Context-Aware Multi-Agent Coordination Framework is proposed that allows the organization of the work of multiple specialized agents to study the context of the user and assess service providers in cooperation with one another. The agent interactions are coordinated by a central coordination module, and the agents that analyze the context, service retrieval and distance evaluation and availability verification compile the relevant information about the potential service providers. The gathered information is then subjected to adaptive ranking system that dynamically reconfigures decision parameters based on the user so that more precise and reliable service suggestions can be provided. It is proposed that the framework is valuable to the research field of Artificial Intelligence and Intelligent Systems, specifically, in the multi-agent systems, situational computing, and intelligent service recommendation domain. Additionally, the system is in line with SDG number 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities

and Communities) as it enhances smart digital services and accessibility and efficiency of city service discovery systems.

1.2. Literature Review

The latest developments in artificial intelligence have been of great help in enhancing intelligent service discovery systems using technologies like natural language processing, voice interaction, and context-based recommendation models. Contemporary service platforms operate under AI-based mechanisms that capture user requests, service intentions and provide recommendations based on automated decision-making systems. Such systems usually combine elements like voice or text input processing, intent detection modules and recommendation engines. As much as these architectures have contributed to increased accessibility and automation, most of the current systems continue to find it difficult to include dynamic contextualization aspects like user location, urgency, travelability, and provider availability in the generation of recommendations. With the dynamic nature of service environments, there is a need to have smarter coordination mechanisms that can handle multiple decision-making processes at any given time and deliver correct context-sensitive recommendations. Multi-agent systems have been recently examined as a promising way of enhancing decision-making under complex settings. Multi-agent architecture is made up of numerous special agents that work together to accomplish various activities like data retrieval, analysis and evaluation. These systems can enhance scalability, adaptability and overall system performance relative to traditional centralized methods by splitting tasks among autonomous agents and having a central coordinator to control and allocate tasks. This distributed design permits more adaptable problem solving, especially in real-time system where the information keeps on changing. Several multi-agent systems based on collaboration in solving complex decisions have been suggested. These platforms use special agents which facilitate, analyze and assess solutions in terms of mutual interactions. These strategies prove that the efficiency of the system can be enhanced because of coordinated collaboration of the agents since they can analyze various facets of a task simultaneously. Most of these systems, however, are mainly concerned with decomposing the tasks and the collaboration of agents and less support given to the potential of incorporating real-time context in the overall decision-making process. The current service recommendation systems are mainly based on rule-based filters, centralized ranking functions, or fixed scoring components when determining the right service providers. These methods tend to rank providers in terms of user ratings, service types or pre-determined place filters. They are extensively applied to commercial service platforms, but they do not serve well to take into account dynamic contextual factors like traffic conditions, real-time service availability, response time, or urgency of the user request. Accordingly, the suggested services might not necessarily be the most feasible or efficient services. Most digital services platforms are also commercial in nature and support automated service discovery services, which enable customers to find providers in their locality. These sites are, however, mainly dependent on a static ranking technique and search mechanism which depends on keywords. Their provider evaluation procedures tend to assess providers based on the fixed criteria without a dynamically orchestrated multiplicity of contextual factors. Also, service discovery in those systems is typically approached as a one-order decision making process instead of a

coordinated process of multiple specialized modules. The other major shortcoming that has been seen in most of the multi-agent systems in place today is that there are no adaptive coordination systems that adapt dynamically in response to user context regarding the criteria used in decisions. In most scenarios, the agents undertake autonomous functions yet fail to put the contextual knowledge into consideration when making the final suggestion. In addition, conventional ranking models usually use predetermined weighting methods that cannot adjust to altered conditions such as the emergency requests or time-sensitive services. The shortcomings point to a major gap in research when it comes to creating intelligent coordination structures that can not only incorporate various contextual factors but also organize the work of the specialized agents. To fill this gap, the suggested Adaptive Context-Aware Multi-Agent Coordination Framework provides a centralized coordination scheme that regulates the relations between the specialized agents, which analyze the situation, find services, estimate range and availability, and rank the providers. The adaptive scoring system is a dynamic system that adapts the ranking criteria based on the context of the user and therefore can provide more accurate and reliable recommendations of the service in the dynamic environment.

1.3.Research Gap

Though recent studies have focused on the application of multi-agent systems and intelligent recommendation models in complex decision-making tasks, there are still a number of limitations being found in the current approaches when applied to the real-time service discovery setting. Numerous existing service recommendation systems are based on the centralized ranking algorithms model or rule-based filtering-tools, in which service providers are rated by a set of fixed criteria or category similarity or location. Although these procedures offer a fundamental service that matches their capabilities, the methods do not provide a dynamic ability to conform to the evolving contextual conditions like urgency of the user, real-time availability of providers, travel viability, and response time. Consequently, the propositions created by such systems might not be reflective of the most appropriate service provider in real-life situations where the circumstances require a lot of changes. Moreover, despite the introduction of multi-agent architectures to enhance scalability of systems and distributed decision-making, most of the current multi-agent systems are based on task decomposition or collaborative reasoning as opposed to adaptive coordination. In most applications, the actual agents are autonomous to carry out their respective specialized functions (information recovery or data mining) yet the tools that coordinate agent responses tend to be fixed. This drawback makes the system pressurized; it is not capable of dynamically changing the decision strategies according to the change in context during the recommendation process. The other weakness that has been witnessed in the existing practices is the application of fixed or fixed ranking processes when assessing service providers. The conventional ranking models usually utilize unchangeable weighting policies in integrating the various assessment parameters. But, in the real-world service requests, they tend to have different contextual priorities. As an example, in cases of emergency service requests, priority of response time and distance might be a requirement whereas in non-emergency cases, priority might be based on reputation of the provider or service quality. The fixed ranking models cannot adjust appropriately to these contextual differences in a dynamic way resulting to poor service recommendations. These shortcomings underscore the existence of a gap in current research in the establishment of adaptive coordination structures that can dynamically combine the contextual information with collaborative multi-agent decision-making. Intelligent coordination mechanisms, capable of organizing several specialized agents, and at the same

time adapting the ranking strategies to user context at the same time environmental conditions are lacking. To bridge this gap the given research suggests Adaptive Context-Aware Multi-Agent Coordination Framework which suggests the introduction of a centralized coordination mechanism in charge of the interactions between specialized agents that conduct context-analysis, service-retrieval, distance-evaluation, availability-check and service-ranking. Contrary to the traditional systems, the proposed framework will include an adaptive scoring system that dynamically adjusts the ranking weights under the conditions of the context, including the urgency level, location restrictions, and availability of providers. This is an adaptive coordination strategy which helps the system generate more accurate context-sensitive recommendations of service to enhance the effectiveness and responsiveness of intelligent service discovery systems.

Feature / Capability	**Rule-Based Service Matching Systems**	**Traditional Recommendation Systems**	**Existing Multi-Agent Systems**	**Proposed Adaptive Multi-Agent Coordination Framework**
Context Awareness	No	Limited	Partial	**Yes**
Dynamic Ranking Adjustment	No	NO	Limited	**Yes**
Real-Time Provider Availability Analysis	Limited	Limited	Partial	**Yes**
Multi-Agent Collaboration	No	No	Yes	**Yes**
Adaptive Decision Making	No	No	Limited	**Yes**
Integration of Multiple Context Factors	Limited	Partial	Limited	**Yes**
Intelligent Service Provider Ranking	Basic	Moderate	Moderate	**Advanced**
Real-Time Recommendation Adaptation	No	Limited	Limited	**Yes**

Table 3 : 1.3.Research Gap

2. OBJECTIVE

2.1.Main Objective

To architect and execute an Adaptive Context-Aware Multi-Agent Coordination Framework to dynamically coordinate specialized agents and utilize an adaptive scoring mechanism to select and propose the most appropriate service providers depending on contextual factors including location, urgency, provider availability and approximate response time.

2.2. Specific Objectives

- To develop a Multi-Agent Coordination module where service discovery activities are shared by specialized agents and coordination between them is managed to facilitate collaborative decision-making.
- To create a Context Analysis Agent which detects and processes contextual information like the location of the user, request time and urgency level to provide context-sensitive recommendations of services. Service Retrieval and Evaluation Agents are to be used to retrieve the possible service providers and consider the aspects of distance, estimated time, and availability of service providers.
- To develop a Ranking Agent to have an adaptive scoring model that will rank the service providers available in the market by considering various factors such as provider rating, reliability, speed of response, and estimated arrival time.
- To apply a context-adaptive weighting system that will dynamically adapt ranking criteria depending on various user conditions like an emergency or non-urgent service request. To determine the effectiveness of the proposed framework in determining the accuracy and efficiency of service recommendations in the dynamic service discovery context.

3. METHODOLOGY

3.1. Research Approach

This study will take on the Design Science Research (DSR) approach with experimental assessment in developing, testing and assessing the proposed Adaptive context-aware Multi-agent Coordination Framework. Design Science Research is appropriate in this research since the main aim is to create an innovative product, in this case an intelligent coordination framework to overcome the limitations in the current service discovery system.

The study starts with the identification of the deficiencies of the existing service recommendation procedures, namely the absence of adaptive coordination and contextual sensing. A suggested framework will be based on these gaps, which will form a multi-agent one in which specialized agents will cooperate to interpret contextual variables and assess service providers. The system will be then adopted as a prototype system to show the coordination mechanism and adaptive ranking model. Lastly, there will be some experimental assessment of how the suggested framework will be effective in coming up with correct and environmentally sensitive service recommendations.

3.2. Technical Methods and Algorithms

The proposed component is realized through multi-agent coordination architecture of a central coordination module and several specific agents that will perform different tasks in the service discovery process.

Multi-Agent Coordination Mechanism

The Multi-Agent Coordinator is the controller of the system. It takes processed user request in the intent detecting module and allocates tasks to specialized agents. The coordinator is also in charge of communication among the agents as well as the combination of their results in order to produce the final recommendation.

Context Analysis Method

The **Context Agent** analyzes contextual information including:

- User location
- Request time
- Urgency level

It is based on these contextual characteristics that the significance of various evaluation criteria is identified in ranking the process.

Service Retrieval Algorithm

Filtering techniques on the service database are used to retrieve candidate service providers to the Search Agent based on:

- Service category

- Geographic location
- Service specialization

The step will give a list of possible providers of relevance to the request by the user. Calculation of Distance and Travel Time.

The Distance Agent determines the feasibility between the user and service providers based on geospatial calculations. Distance and estimated time of travel are calculated to establish the feasibility of each provider reaching the user.

Availability Verification

To determine the availability of the service providers, Availability Agent looks at:

- Online status
- Current workload
- Job acceptance behavior

Unavailable and inactive providers are eliminated off the list.

Adaptive Ranking Algorithm

Remaining providers are judged by the Ranking Agent based on an adaptive scoring model. Every provider will be rated according to several assessment criteria that include:

- Provider rating
- Reliability
- Response speed
- Estimated arrival time
- Availability

The adaptive scoring formula may be formulated as:

Final Score = Rating w1 + Reliability w2 + Response Speed w3 + Availability w4 +ETA Score w5.

The weights w1-w5 are dynamically changed depending on some contextual factors. As an example, during an emergency the system assigns more weight to travel time and availability whereas when the situation is not urgent, more weight is given to the reputation of the provider as well as the reliability of the provider.

3.3. Tools and Platforms

The proposed framework will be implemented using the current AI and software development technologies in order to guarantee scalability and effective coordination of agents.

Programming Language – Python

As the main programming language, Python is going to be utilized because of the large-scale support of artificial intelligence, machine learning, and agent-oriented structures.

Backend Framework – FastAPI / Flask

To ensure that the agents communicate with each other and requests of the service are processed, a lightweight framework like FastAPI or Flask will be employed.

Database – MongoDB / PostgreSQL

The information of service providers, history of services and the contextual data needed in evaluating recommendations will be stored in a database system.

Geospatial Processing – Google Maps API / OpenStreetMap

Distance calculations, travelling time, and estimated arrival time between users and service providers will be calculated using the geospatial APIs.

Development Platform – Visual Studio Code

The development environment will be Visual Studio Code because it has a great deal of support of Python development and debugging tools.

These technologies are chosen due to their flexibility, scalability and good integration potential to put up multi-agent AI systems.

3.4. Data Collection and Ethical Considerations

The analysis of the suggested framework needs information concerning the service providers and requests. The data will be gathered based on simulated service request conditions, and publicly accessible datasets connected to the characteristic of service providers like rating, availability, and location.

Synthetic datasets will be created to act as a simulator of real-world service requests in various contextual conditions such as emergency requests, normal requests as well as different geographic constraints. The technique is useful to experiment with things without revealing sensitive user information.

In terms of ethical considerations, the following will be done:

- Personally identifiable user data is not stored or collected.
- All the experimental data are simulated or anonymized.
- There is institutional research in terms of data collection.

Also, the design of the system will be without promoting favoritism and transparency in the selection of providers using biased ranking strategies.

3.5. Validation Strategy and Performance Metrics

The viability of the suggested Adaptive Context-Aware Multi-Agent Coordination Framework will be tested with the help of the experimental testing based on the simulated situations of service requests.

The system will be put into test in varying contextual conditions such as varying levels of urgency, location limitations and availability of service providers.

The metrics of performance that will be used to assess the framework are as follows:

Recommendation Accuracy

Determines the effectiveness of the system in choosing the best service provider, based on the conditions in the set context.

Response Time

Time taken to analyze, based on contextual factors and produce a service recommendation is measured.

System Adaptability

Assesses the performance of the ranking mechanism to modify its decision criteria to fit various contextual conditions including in case of an emergency or a normal request.

Provider Selection Efficiency

Determines how well the system can filter and rank the service providers and remove unavailable or impractical ones.

These experiments will be examined to find out whether the presented framework enhances the effectiveness and the flexibility of service recommendations in comparison to traditional fixed recommendation strategies.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1.Overview of Proposed Solution and Its Components

The solution hypothesized is a multilingual voice-based local service discovery system with behavior-awareness, which helps users to identify an appropriate local service provider using intelligent and adaptive decision-making operations. At the architectural level, the architecture consists of few key elements including user input, speech or text processing, intent detection, the Adaptive Context-Aware Multi-Agent Coordination Framework, and generating final recommendation. Various workflow steps are involved in the overall system where a user request is received, followed by taking the request and identifying the intended service and directing the identified intent to the multi-agent coordination framework to select the provider. As per your project description, the component commences once intent detection has taken place and must organize specialized agents to produce the most appropriate service recommendation. The main decision-making element of this architecture is the Adaptive Context-Aware Multi-Agent Coordination Framework. It is meant to coordinate several specialized agents that examine the situational aspects, access candidate providers, assess travel viability, test provider availability, and rank providers based on an adaptive scoring system. This element enhances the quality of the recommendations, by integrating the distributed agent collaboration with context-based ranking instead of based on the use of static provider filtering or mere location-based matching. The framework then serves to provide the intelligent layer that translates the service detected to one that is practical and reliable service recommendation.



Figure 1 ; 4.1. Overview of Proposed Solution and Its Components

4.2 Highlight the individual component and its parts

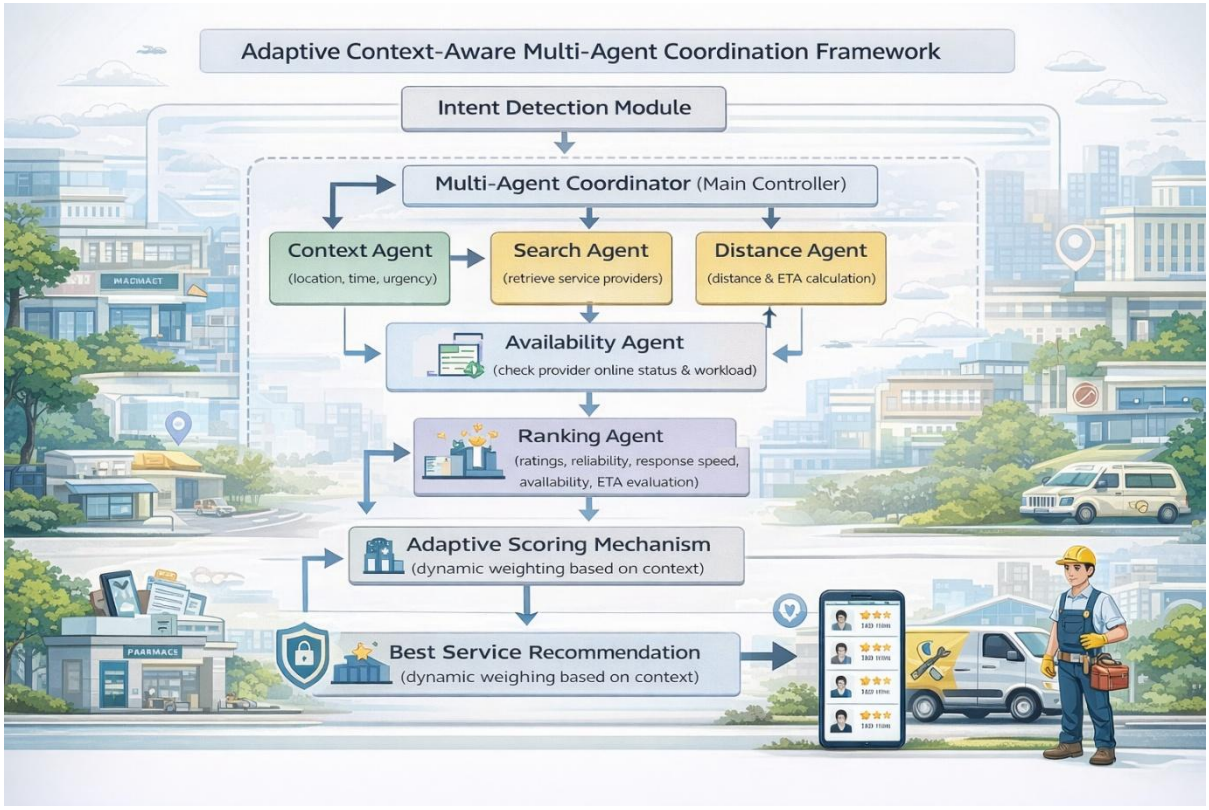


Figure 2 :Highlight the individual component and its parts

The Adaptive Context-Aware Multi-Agent Coordination Framework is the smart decision-making interface of the system. Once the intent detection module has detected the requested service, Multi-Agent Coordinator allocates tasks to specialized agents tasked with the responsibility of analyzing the contextual information, finding candidate providers, determining whether travel is possible, and ensuring that the provider is available.

The agents handle a particular sort of information and send their results to the coordinator. The Ranking Agent will then rate providers based on several factors including ratings, reliability, speed of responding and the estimated arrival time. Lastly, the adaptive scoring system is able to dynamically adjust the weights used in evaluations depending on the current contextual factors like the urgency level, and as such, the system can suggest the most appropriate service provider.

4.3Component Interactions



Figure 3 :Component Interactions

Adaptive Context-Aware Multi-Agent Coordination Framework only starts its work once the Intent Detection Module has recognized the requested service and transmitted the intent to the Multi-Agent Coordinator. The coordinator is the controlling unit of the component and is charged with the task of allocating the tasks to the specialized agents, gathering their results and drawing up the final recommendation. Context Agent, in this context, examines the environmental and user specific parameters, including the location, request time and level of urgency whereas Search Agent retrieves candidate service providers out of the database, based on the service category and location. Meanwhile the Distance Agent assesses the feasibility of travelling distance and the estimated arrival time and the Availability Agent assesses the provider online, free and able to accommodate the request.

When these specialized agents have finished with their assigned tasks, their output is sent to the Ranking Agent. The Ranking Agent assesses the candidate providers based on the criteria which include ratings, reliability, response speed, ETA and availability. The Adaptive Scoring Mechanism then reinforces this interaction by changing the significance of ranking criteria in real-time. The framework, as an example, assigns more priority to ETA and availability in the case of urgent situations whereas in normal cases, the provider rating and reliability may be prioritized more. It is a coordinated interaction through which the system makes sure that final

choice of provider is not made based on one factor but based on a thousand collaborating agent decisions.

4.4 Data Flow Explanation

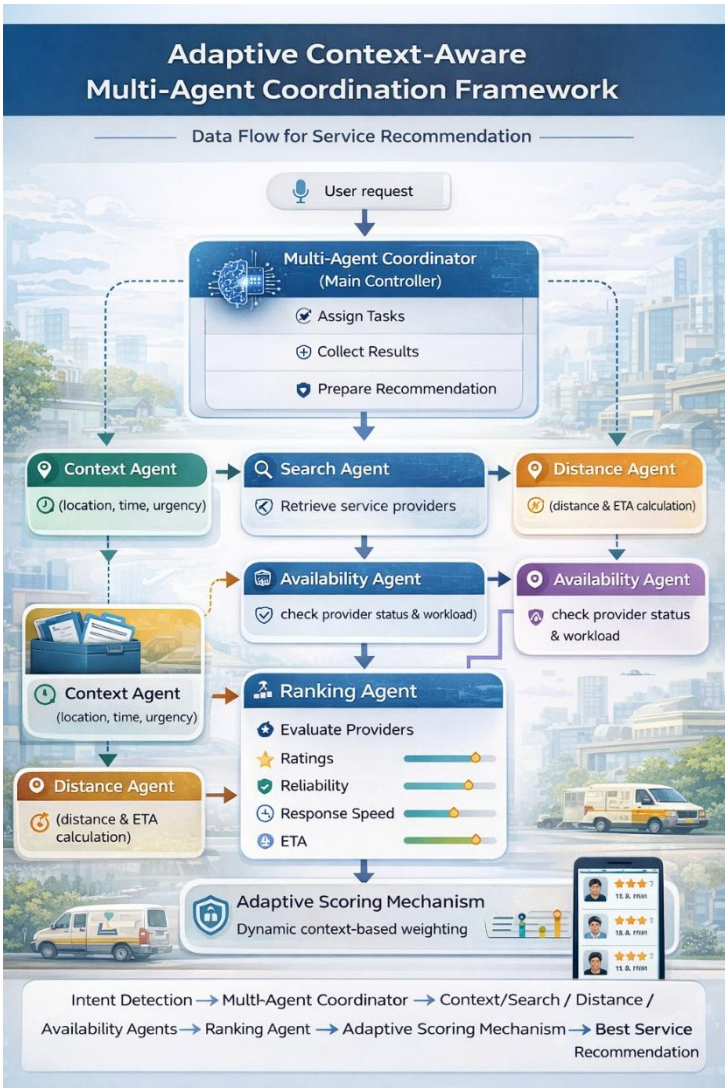


Figure 4 :Data Flow Explanation

The Adaptive Context-Aware Multi-Agent Coordination Framework data flow commences at the point when the Intent Detection Module transmits the identified service request to the Multi-Agent Coordinator. This request is spread by the coordinator to more expert agents such as Context Agent, Search Agent, Distance Agent and Availability Agent. The agents process information e.g. user context, service providers, travel distance and availability of service providers.

The information collected after processing is passed to the Ranking Agent that is used to evaluate providers based on several factors including ratings, reliability, speed of response, estimated arrival time (ETA) and availability. Adaptive Scoring Mechanism, in turn, adaptively modulates the significance of these factors depending on the situational factors like urgency level. The best service provider is then chosen and sent back as the Best Service Recommendation and the data flow of the component is complete.

4.5 Scalability and Security Considerations

The architecture proposed is scalable as it is a multi-agent structure, which is composed of independent agents doing specialized duties. This modular structure enables subsequent agents or other evaluation criteria to be added later without any redesigning of the entire system. As an example, future iterations may have fairness-conscious allocation agents, provider trust estimation agents, or dynamic reassignment agents. Distributed framework also better supports an increase in the type of service, number of requests made by users and services given by the provider than one centralized recommendation module.

Security wise the component should provide secure management of user and provider data particularly location data and availability information. Provider records must be restricted by use of authenticated system modules and inter-system layer communications must be secured to avoid the exposure of unauthorized data. Because the framework is based on multiple interacting modules, the secure API processing and safe database access are significant to ensure the confidentiality of data and system integrity. Moreover, the selection of the providers should not be biased or unfair since the evaluation criteria and ranking logic must be transparent and controlled. This is significant as the coordination system directly affects the result of recommendations and utilizes confidence on the platform.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Adaptive Context-Aware Multi-Agent Coordination Framework involves several stakeholders who interact with or benefit from the system.

Service Users (Customers)

Service users are individuals who request services through the platform. They expect the system to recommend the most suitable service providers based on their location, urgency level, and service requirements.

Service Providers (Employees / Workers)

Service providers are registered professionals such as plumbers, electricians, mechanics, or other workers who offer services through the platform. They depend on the system to receive fair and relevant service requests that match their availability, skills, and location.

System Administrators

System administrators manage the platform, maintain the service provider database, and monitor system performance. They ensure that the coordination framework functions properly and that the system remains reliable and secure.

Platform Developers / Researchers

Developers and researchers are responsible for designing, implementing, and improving the multi-agent coordination framework. They evaluate the system's performance and refine algorithms to improve recommendation accuracy and system efficiency.

5.2 Functional Requirements

Functional requirements describe the **specific operations and capabilities** that the proposed component must perform.

Requirement ID	Functional Requirement Description
FR1	**Receive Service Request Information** The system shall receive processed user requests and service intent information from the intent detection module.
FR2	**Coordinate Multi-Agent Operations** The Multi-Agent Coordinator shall distribute tasks to specialized agents and manage communication between them.
FR3	**Analyze User Context** The Context Agent shall analyze contextual factors such as user location, request time, and urgency level to support context-aware recommendations.
FR4	**Retrieve Candidate Service Providers** The Search Agent shall retrieve relevant service providers from the service provider database based on service category and location.
FR5	**Evaluating Distance and Travel Feasibility** The Distance Agent shall calculate the distance and estimated arrival time between users and service providers.
FR6	**Verify Provider Availability** The Availability Agent shall verify whether service providers are currently online, available, and capable of accepting service requests.
FR7	**Rank Service Providers** The Ranking Agent shall evaluate candidate providers using multiple criteria including ratings, reliability, response speed, availability, and estimated arrival time.
FR8	**Apply Adaptive Scoring Mechanism** The system shall dynamically adjust ranking weights based on contextual factors such as urgency level or service type.

FR9	**Generate Final Service Recommendation** The Multi-Agent Coordinator shall collect output from all agents and generate the final recommendation for the most suitable service provider.

Table 4 :Functional Requirements

5.3 Non-Functional Requirements

Non-functional requirements describe how well the system performs its functions rather than what functions it performs.

System Performance	The coordination framework must generate service recommendations within a short response time to support real-time service requests. This is a non-functional requirement because it defines the **speed and efficiency of the system**, rather than a specific function.
Scalability	The system should be capable of handling increasing numbers of service requests and service providers without performance degradation. This is non-functional because it relates to the **system's ability to maintain performance under increasing workload**, not a direct functional operation.
Reliability	The system must consistently produce accurate and dependable service recommendations even when contextual conditions change dynamically. This requirement is non-functional because it describes the **consistency and dependability of system performance** rather than a specific task.
Security	The system must protect sensitive user and provide data through secure authentication and controlled data access mechanisms. This is a non-functional requirement because it focuses on **protecting system data and preventing unauthorized access**, not on performing a system function.
Usability	The system should support an easy and intuitive interaction process so that users can request services without difficulty. This is non-functional because it concerns the **quality of user interaction and system accessibility**, rather than a specific operational function.
Adaptability	The coordination framework should dynamically adjust ranking criteria according to contextual conditions such as urgency level or service type . This is non-functional because it describes the **system's ability to adapt its behavior under different conditions**, rather than a direct functional task.

Table 5 :Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The proposed system is mainly aimed at people who need to have the opportunity to obtain services that can be accessed easily and offer reliable solutions like plumbing, electrical repairs, mechanical services, healthcare support, and emergency support. The key groups of customers will be people living in cities, businesspersons, and those who often use digital platforms to find service providers around them. Also, another important stakeholder group is small-scale service providers, including plumbers, electricians, technicians, and freelance workers, as the service will allow them to communicate with potential customers and accept requests for their services.

The common customer profile of the site would be a smartphone user with an urban or semi-urban location and needs some immediate help with finding a local or nearby service provider. Such users demand quick, dependable and context-sensitive suggestions of service, which consider aspects like distance, reaction, and service quality. The platform also facilitates the service providers who are interested in equal employment prospects and popularity in the local service markets.

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6.2. Value Proposition

The main value of the proposed system is the fact that it is able to give intelligent and context-based service recommendations through a multi-agent coordination framework. As opposed to the traditional service discovery systems, which are based on the principles of search and ranking being static, the suggested system is realized as a dynamic system that analyzes the contextual factors, including user location, urgency level, the availability of service providers, and the possibility to travel, to suggest the most appropriate service providers.

Such an adaptive coordination system advances the reliability of the services, lessens response period, and increases user satisfaction by making the recommended service providers reliable and capable of responding within a short time among other factors. To the service providers, the system makes the services more visible and plays a role in the distribution of employment opportunities in a more efficient manner.

6.3. Revenue Model

The site has several business models that can be used to generate revenue. A commission-based model is one of the main sources of revenue with a small percentage fee being paid on each successful service transaction that is done through the platform. This strategy will see that the revenue is directly attached to the use of platforms.

Service providers have another source of revenue, which is a subscription-based model. Providers can purchase premium whereby they are given advantageous features like increased exposure in search engine results, priority notifications among other profile features.

Also, the platform can bring advertising features to service providers or businesses that desire to market their services to a more significant number of people.

6.4. Cost Estimation

The process of developing and deploying the platform has a number of cost aspects. The first costs would involve software development, system design, and deployment of multi-agent coordination framework. The rest of the expenses can be related to cloud hosting solutions, database architecture, and the use of geospatial API to calculate distances and travel time.

The operational costs will consist of the maintenance of the system, hosting of a server, technical support and constant system enhancement. The marketing costs can also be necessary to publicize the platform and appeal to users as well as providers of service services.

6.5. Pricing Strategy

The pricing will be aimed at ensuring that it is affordable to the users but sustainable to the platform. A minimal commission fee can be imposed on any completed service transaction to be able to produce revenue but without a huge weight on service costs to the user.

As a service provider, we can also offer flexible subscription plans so that the provider has an option of free basic accounts and premium accounts with extra features like higher ranking or a more efficient analytics service. Such a tiered pricing approach will allow accessibility and generate revenue opportunities to the platform.

6.6. Competitive Advantage

The suggested system has a number of competitive advantages over the conventional service discovery platforms. Most of the available platforms are based on the fixed ranking algorithms and the basic location search options which is frequently inability to take into account dynamic contextual variables.

An Adaptive Context-Aware Multi-Agent Coordination Framework provides flexibility to the system to combine analysis of various conditions in the context and dynamically change rankings of service providers. Such a smart coordination system enhances the accuracy of the recommendations and allows responding to urgent situations of service faster.

Moreover, multi-agent architecture can improve the scalability and flexibility of the system, which provides the platform to effectively operate in complex service environments and grow the number of service providers.

6.7. Intellectual Property Considerations

The suggested mechanism of coordination and adaptive ranking can be registered as an intellectual property. The coordination logic between specialized agents, system architecture, and adaptive scoring algorithms are new technical contributions which may be potentially safeguarded by software copyright or patent application.

Securing the intellectual property of the framework will help in ensuring that the technology developed will be unique and will not be duplicated unlawfully by the competitors. After having a system design, implementing proper documentation of the algorithms, architecture will assist in future intellectual property registration in case the platform is commercialized.

7. BUDGET AND JUSTIFICATION

The total estimated budget for developing and evaluating the **Adaptive Context-Aware Multi-Agent Coordination Framework** is **60,000 LKR**, which falls within the acceptable research budget range. Most of the budget is allocated to cloud services and hardware upgrades required to support the execution of multiple agents and real-time service recommendation experiments. The use of open-source software tools helps significantly reduce overall development costs while maintaining system performance and scalability.

7.1 Hardware Costs

Hardware Item	Description	Estimated Cost (LKR)
External SSD / Storage Device	Used to store service datasets, system logs, and experimental results during testing of the multi-agent framework	6,000
GPU Upgrade / High-performance workstation access	Improves system performance when running multiple agents, database operations, and API requests simultaneously	10,000
Backup Storage (USB / SSD)	Used for maintaining backup copies of datasets and project files to prevent data loss	4,000
Total Hardware Cost		**20,000 LKR**

Table 6 :Hardware Costs

Justification

The proposed component involves running several agents simultaneously, including context analysis, service retrieval, and ranking agents. These processes require sufficient memory and storage capacity. Additional RAM and external storage improve development efficiency and ensure reliable storage of experimental data generated during testing.

7.2 Software Licenses

Software Tool	Purpose	Estimated Cost (LKR)
Python Development Environment	Implementation of multi-agent coordination algorithms	Free
Visual Studio Code	Development and debugging of system modules	Free
AI Libraries (NumPy, Pandas, Scikit-learn)	Data processing and algorithm development	Free
MongoDB / PostgreSQL Community Edition	Database for storing service provider and contextual data	Free
Total Hardware Cost		**Free**

Table 7 : Software Licenses

Justification

The system will be developed entirely using **open-source software technologies**, which eliminates licensing costs while still providing powerful tools for implementing AI-based multi-agent systems.

7.3 Cloud Services

Cloud Resource	Purpose	Estimated Cost (LKR)
Cloud Server Hosting (6–8 months)	Deploying and testing the multi-agent coordination framework in a simulated real-time environment	12,000
Cloud Database Hosting	Storing service provider data and contextual information used by the agents	4,000
Geospatial API Usage (Google Maps / OpenStreetMap)	Distance calculation and ETA estimation used by the Distance Agent	8,000
Total Cloud Services Cost		24,000 LKR

Table 8 : Cloud Services

Justification

Cloud infrastructure is necessary to host the prototype system and test real-time interactions between agents. The geospatial API is essential for calculating distances and travel times between users and service providers, which directly supports the functionality of the Distance Agent in the coordination framework.

7.4 Human Effort Cost

Activity	Description	Estimated Cost (LKR)
Dataset Collection	Gathering and organizing service provider information for testing the system	5,000
System Development	Designing and implementing the multi-agent coordination framework	6,000
Testing and Evaluation	Running experiments and measuring system performance	5,000
Total Human Effort Cost		16,000 LKR

Table 9 :Human Effort Cost

Justification:
Human effort is required to collect relevant service provider datasets, implement system modules, integrate agent communication mechanisms, and evaluate system performance under different service request scenarios. These activities are essential for validating the effectiveness of the proposed coordination framework.

7.5 Total Estimated Budget

Category	Cost (LKR)
Hardware Costs	20,000
Software Licenses	Free
Cloud Services	24,000
Human Effort Cost	16,000
Total Budget	**60,000 LKR**

Table 10 : Total Estimated Budget

Justification

The total estimated budget of **60,000 LKR** is considered sufficient to support the development and evaluation of the **Adaptive Context-Aware Multi-Agent Coordination Framework**. The budget mainly covers essential resources such as hardware improvements for running multiple agents efficiently, cloud infrastructure for deploying and testing the prototype system, geospatial API usage for distance and travel time calculations, and limited human effort required for dataset preparation and system testing. The use of open-source software tools significantly reduces development costs, ensuring that the project remains financially feasible while still providing the necessary resources to implement and validate the proposed multi-agent coordination framework effectively.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Task ID	Task Name	Description	Timeline
T1	System Architecture Design	Design the architecture of the Adaptive Context-Aware Multi-Agent Coordination Framework including agent roles and coordination workflow.	March 2026
T2	Context Analysis Module Development	Implement the Context Agent to analyze contextual information such as user location, request time, and urgency level.	April 2026
T3	Service Retrieval Agent Development	Develop the Search Agent to retrieve candidate service providers from the database based on service category and location.	April 2026
T4	Distance Evaluation Module	Implement the Distance Agent to calculate travel distance and estimated arrival time using geospatial APIs.	May 2026
T5	Availability Verification Module	Develop the Availability Agent to check service provider availability and filter inactive providers.	May 2026
T6	Adaptive Ranking Algorithm Development	Design and implement the adaptive scoring model that ranks service providers based on contextual conditions.	June 2026
T7	Multi-Agent Coordination Integration	Integrate all agents through the central coordination module to enable collaborative decision making.	June – July 2026
T8	System Testing and Validation	Evaluate recommendation accuracy, response time, and system performance under different contextual scenarios.	July – August 2026
T9	Final System Optimization	Improve algorithm performance and prepare the component for final system integration and report submission.	September 2026

8.2 Timeline Alignment

All the WBS activities of the Adaptive Context-Aware Multi-Agent Coordination Framework are scheduled in accordance with the official RP project timeline, so that development activities do not hinder such a key academic milestone as the submission of the proposal, the presentation of the progress, and the final review. System design and agent module development are undertaken at the early stages whereas system integration, testing and validation are undertaken at the later stages. This organized alignment would mean that the component is moving smoothly and is prepared to be demonstrated in the necessary project evaluations.

8.3 Balanced Workload Distribution

The Adaptive Context-Aware Multi-Agent Coordination Framework development tasks are also split between the various project stages to create a balanced workload during the research time frame. The initial phases are concerned with system architecture design and initial module development, whereas

the third stage is concerned with the implementation of the main core agents including the context analysis, service retrieval and adaptive ranking. The last stage focuses on system integration, testing and performance validation to confirm that the framework works to the final test before the final evaluation. Such even distribution of activities assists in the preservation of constant progress and elimination of the overload at a given phase of the project. It also makes sure that enough time is provided both in the development process as well as evaluation process and that before the final submission and presentation, the component is refined and validated.

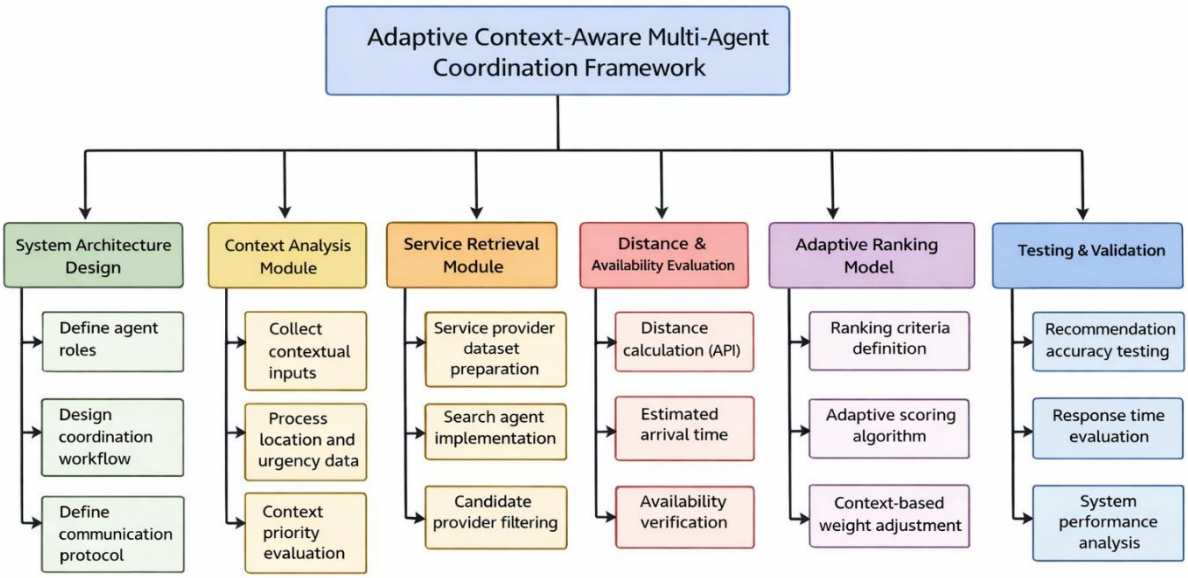


Figure 5 : Adaptive Context-Aware Multi-Agent Coordination Framework

9. GANTT CHART

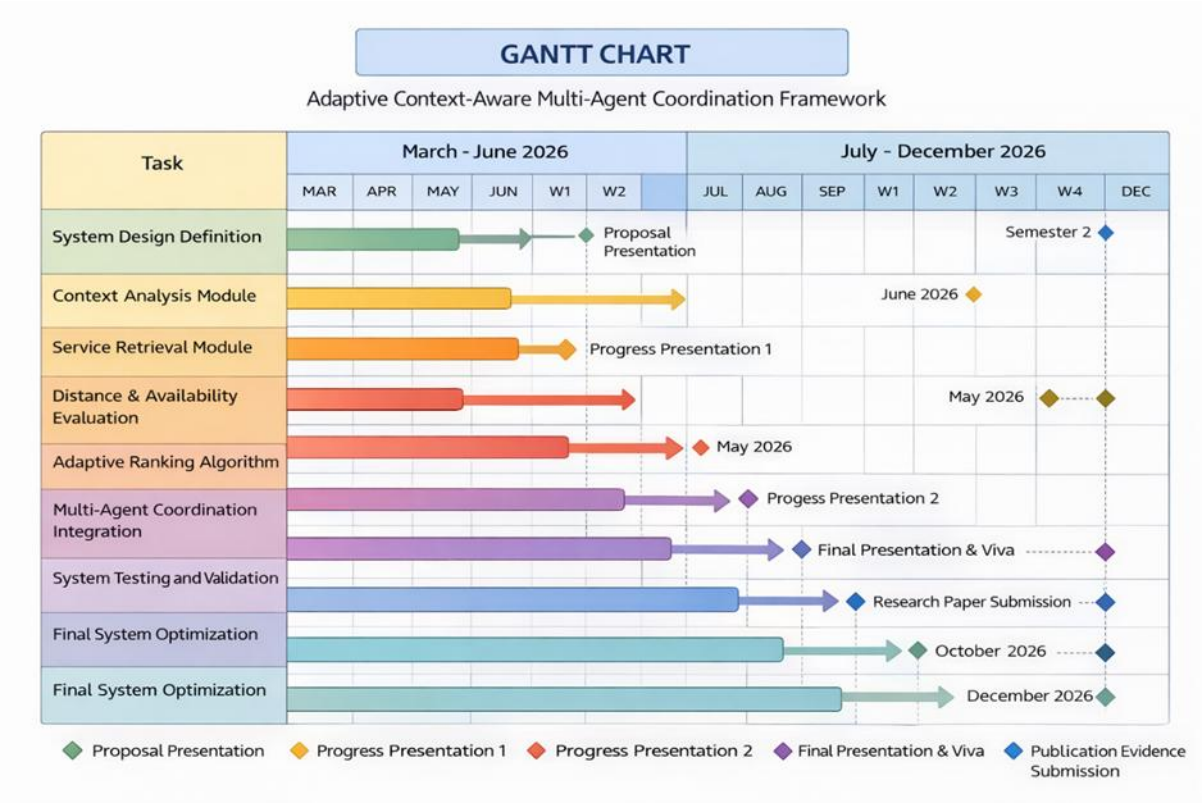


Figure 6 : GANTT CHART

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11.APPENDICES

11.2. Appendix A – Survey Instruments

he initial survey will be carried out to know the user behavior and need as far as the digital service discovery platforms are concerned. The survey will gather data on the current trends in terms of local services search by the users, the difficulties they experience in choosing the credible service providers and their expectations of intelligent service recommendation systems.

Some of the questions that will be in the questionnaire will be the urgency of the service, the importance of the ratings of the provider, consideration of travelling time, and communication methods that would be preferred when ordering the services. The obtained responses will be used to justify the necessity of a context-aware recommendation mechanism and guide the design of Adaptive Context-Aware Multi-Agents Coordination Framework..

11.3Appendix B – Detailed System Diagrams

This appendix includes the technical diagrams that describe the system architecture and the internal workflow of the proposed component. The diagrams illustrate how the **Multi-Agent Coordinator interacts with specialized agents such as Context Agent, Search Agent, Distance Agent, Availability Agent, and Ranking Agent** to generate service recommendations.

These diagrams help explain the coordination mechanism, agent communication flow, and decision-making process within the system. They also demonstrate how contextual information is analyzed and integrated into the adaptive scoring mechanism to improve service recommendation accuracy

Appendix C – Preliminary UI Mockups

Preliminary user interface mockups are designed to visualize how users will interact with the service discovery platform. The mockups include screens such as the service request interface, service recommendation results page, and service provider profile view.

These UI designs demonstrate how users can easily submit service requests and view recommended service providers based on contextual factors such as location, availability, and estimated response time. The mockups also help ensure that the platform remains user-friendly and accessible for a wide range of users.

Appendix D – Risk Analysis

Several potential risks may affect the development and implementation of the proposed system. One potential risk is the limited availability of real-world service provider datasets, which may impact the accuracy of system evaluation. Another risk is dependency on external APIs for geospatial calculations, which may introduce limitations in terms of usage quotas or service reliability.

To mitigate these risks, alternative datasets and backup geospatial services can be considered. Additionally, simulated service scenarios can be used during system testing to evaluate the coordination framework under controlled conditions.

Appendix E – Ethical Documents

The research will follow ethical guidelines to ensure responsible data usage and protection of user privacy. No personally identifiable user data will be collected during system evaluation. Any datasets used for experimental testing will be anonymized or synthetically generated to prevent privacy violations.

Furthermore, the system will be designed to ensure fairness in service provider selection by avoiding biased ranking strategies. Ethical approval documentation and consent procedures will be maintained if any user feedback or survey data is collected during the research process.